

ACTIVITY 14 — FACILITATOR GUIDE

Descriptive Statistics and Implementation Planning

TYPE	DURATION	DMAIC PHASE	MAPS TO
Elective Lab	45 min	CONTROL	A4, A5

Purpose of this activity

Summarise data numerically and plan the rollout (A4, A5).

Close the loop. Compute the measures of central tendency and dispersion that describe your data, then convert the selected countermeasures into a dated implementation plan with owners and barriers identified.

What learners produce

A descriptive statistics summary explaining the shape of the data and a dated implementation plan with owners and barriers.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- A laptop with Excel, Google Sheets or LibreOffice Calc
- The CSV files in this activity's data/ and templates/ folders
- Flipchart or A3 sheet and markers
- The course slides for the scenario and the framework

Set-up

Groups of 3-5. Distribute one worksheet per learner and one checklist per group. Every activity runs on the same continuous Contoso Service Desk scenario, so remind learners where this one sits in that story before starting.

Scenario / briefing material

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55 minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

Files in this activity folder

data/ — the mock data you analyse:

- assignment-times.csv — 400 rows (date, ticket_id, assignment_time_min)
- daily-summary.csv — 10 rows (date, tickets, mean_assignment_min, median_assignment_min, defects, missed_30min_goal)
- service-desk-tickets.csv — 400 rows (ticket_id, date, ticket_type, channel, assigned_queue, agent, assignment_time_min, met_30min_goal, defect_count, defect_categories, reopened)

templates/ — the worksheets you complete:

- descriptive-statistics.csv (statistic, formula_or_function, your_value)
- implementation-plan.csv (action, owner, start_date, due_date, barrier, mitigation)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

data/assignment-times.csv is the raw column for the statistics.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Compute the mean and median for your assignment-time data and compare them.	Mean 55 vs median 46 is a real right skew — the long tail of late tickets causes it.
2	Compute the range and standard deviation to describe the spread.	Reporting the mean alone on skewed data is the trap; make them say which they would use.
3	Explain what the mean-versus-median difference reveals about outliers and skew.	
4	Write the implementation plan: action, owner, date, barriers and mitigation.	

Suggested timing

Minutes	Phase
10 min	Compute mean and median
10 min	Compute range and standard deviation
10 min	Explain the mean-vs-median gap
15 min	Write the dated implementation plan

Total: 45 minutes

Discussion & decision prompts

1. Why do the mean and median differ in this data, and which would you report?
2. What does the standard deviation tell you that the mean cannot?
3. Which implementation action is most likely to slip, and what is the mitigation?

Debrief

Bring the class back together and draw out the learning:

- Ask two or three groups to show what they produced, not just what they concluded.
- Ask what surprised them — the gap between what they assumed and what the data showed.

- Link the activity explicitly back to the CONTROL phase of DMAIC.
- Remind learners this activity rehearses what the Case Study assessment will require.

Reflection questions:

- Where does your workplace report an average that hides a skew?
- What barrier most often stops improvements landing in your team?

Facilitator notes — what to watch for

- Mean 55 vs median 46 is a real right skew — the long tail of late tickets causes it.
- Reporting the mean alone on skewed data is the trap; make them say which they would use.

Success criterion

You can explain why the mean and median differ in your data, and every implementation action has an owner and a date.

Assessment link

This activity builds the CONTROL phase of DMAIC. All activities are in-class activities assessed indirectly through the open-book Case Study (CS), which reuses the same techniques on the same Contoso Service Desk scenario and the same figures (400 tickets, yield 76%, DPMO 50,000). Knowledge is assessed in the Written Assessment (WA-SAQ). Nothing is assessed that is not practised here or taught in the slides.